



Stony Brook Institute
at Anhui University

安徽大学纽约石溪学院

ISE 333: User Interface Development (Design)

Practice 3: UI-Oriented Computational Geometry

Spring 2026

Instructor: Hao Li

Distance

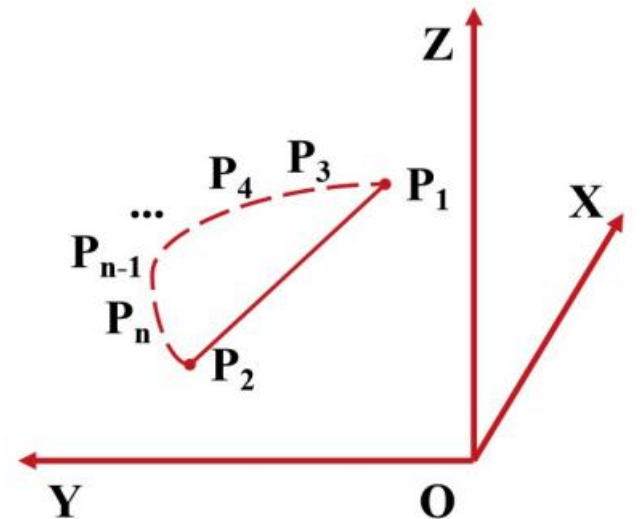
- Euclidean Distance

$$\|\mathbf{P}_1 - \mathbf{P}_2\|_2 = \sqrt{(x_{1,1} - x_{2,1})^2 + (x_{1,2} - x_{2,2})^2 + \cdots + (x_{1,n} - x_{2,n})^2}$$

$$\|\mathbf{P}_1 - \mathbf{P}_2\|_2 = \sqrt{\sum_{i=1}^n (x_{1,i} - x_{2,i})^2} = \sqrt{(\mathbf{P}_1 - \mathbf{P}_2)^T (\mathbf{P}_1 - \mathbf{P}_2)}.$$

- Triangular Inequality

$$\|\mathbf{P}_1 - \mathbf{P}_2\| \leq \|\mathbf{P}_1 - \mathbf{P}_3\| + \|\mathbf{P}_3 - \mathbf{P}_2\|$$



Distance

- Point-To-Line Distance

Given three generic points $\mathbf{P}_1$, $\mathbf{P}_2$, and $\mathbf{P}$ in a n -dimensional vector space

$$\mathbf{P}_1 = \begin{bmatrix} x_{1,1} \\ x_{1,2} \\ \vdots \\ x_{1,n} \end{bmatrix}, \quad \mathbf{P}_2 = \begin{bmatrix} x_{2,1} \\ x_{2,2} \\ \vdots \\ x_{2,n} \end{bmatrix}, \quad \mathbf{P} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

$$\mathbf{P}(t) = t\mathbf{P}_1 + (1-t)\mathbf{P}_2 = (\mathbf{P}_1 - \mathbf{P}_2)t + \mathbf{P}_2.$$

$$\begin{aligned} d &= \min \|\mathbf{P}(t) - \mathbf{P}\|_2 = \min \|(\mathbf{P}_1 - \mathbf{P}_2)t + \mathbf{P}_2 - \mathbf{P}\|_2 \\ &= \min \sqrt{((\mathbf{P}_1 - \mathbf{P}_2)t + \mathbf{P}_2 - \mathbf{P})^T((\mathbf{P}_1 - \mathbf{P}_2)t + \mathbf{P}_2 - \mathbf{P})}. \end{aligned}$$

$$\mathbf{P}_{proj} = \frac{(\mathbf{P}_1 - \mathbf{P}_2)^T(\mathbf{P} - \mathbf{P}_2)}{(\mathbf{P}_1 - \mathbf{P}_2)^T(\mathbf{P}_1 - \mathbf{P}_2)}(\mathbf{P}_1 - \mathbf{P}_2) + \mathbf{P}_2.$$

$$\begin{aligned} t &= \arg \min_t ((\mathbf{P}_1 - \mathbf{P}_2)t + \mathbf{P}_2 - \mathbf{P})^T((\mathbf{P}_1 - \mathbf{P}_2)t + \mathbf{P}_2 - \mathbf{P}) \\ &= \arg \min_t (\mathbf{P}_1 - \mathbf{P}_2)^T(\mathbf{P}_1 - \mathbf{P}_2)t^2 + 2(\mathbf{P}_1 - \mathbf{P}_2)^T(\mathbf{P}_2 - \mathbf{P})t + (\mathbf{P}_2 - \mathbf{P})^T(\mathbf{P}_2 - \mathbf{P}) \\ &= \frac{(\mathbf{P}_1 - \mathbf{P}_2)^T(\mathbf{P} - \mathbf{P}_2)}{(\mathbf{P}_1 - \mathbf{P}_2)^T(\mathbf{P}_1 - \mathbf{P}_2)}. \end{aligned}$$

$$d = \sqrt{\frac{(\mathbf{P}_1 - \mathbf{P}_2)^T(\mathbf{P}_1 - \mathbf{P}_2)(\mathbf{P} - \mathbf{P}_2)^T(\mathbf{P} - \mathbf{P}_2) - ((\mathbf{P}_1 - \mathbf{P}_2)^T(\mathbf{P} - \mathbf{P}_2))^2}{(\mathbf{P}_1 - \mathbf{P}_2)^T(\mathbf{P}_1 - \mathbf{P}_2)}}.$$

- Practice (Class Live Example)

Distance

- Point-To-Plane Distance

Besides the point-to-point distance, the point-to-line distance, how to compute the point-to-plane distance? Given $n + 1$ generic points $\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_n$, and $\mathbf{P}$ in a n -dimensional vector space

$$\mathbf{P}_1 = \begin{bmatrix} x_{1,1} \\ x_{1,2} \\ \vdots \\ x_{1,n} \end{bmatrix}, \quad \mathbf{P}_2 = \begin{bmatrix} x_{2,1} \\ x_{2,2} \\ \vdots \\ x_{2,n} \end{bmatrix}, \quad \dots, \quad \mathbf{P}_n = \begin{bmatrix} x_{n,1} \\ x_{n,2} \\ \vdots \\ x_{n,n} \end{bmatrix}, \quad \mathbf{P} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

View the n points $\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_n$ as n vectors and suppose they are linearly independent, namely suppose the compound matrix formed by $\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_n$

$$\mathbf{C} \equiv [\mathbf{P}_1 \quad \mathbf{P}_2 \quad \dots \quad \mathbf{P}_n] = \begin{bmatrix} x_{1,1} & x_{2,1} & \dots & x_{n,1} \\ x_{1,2} & x_{2,2} & \dots & x_{n,2} \\ \vdots & \vdots & \ddots & \vdots \\ x_{1,n} & x_{2,n} & \dots & x_{n,n} \end{bmatrix}$$

is invertible.

Distance

- Point-To-Plane Distance

Any point on the plane spanned by $\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_n$ can be parametrized as

$$\mathbf{P}(\mathbf{t}) = \mathbf{C} \mathbf{t} \equiv [\mathbf{P}_1 \ \mathbf{P}_2 \ \cdots \ \mathbf{P}_n] \mathbf{t} \equiv \begin{bmatrix} x_{1,1} & x_{2,1} & \cdots & x_{n,1} \\ x_{1,2} & x_{2,2} & \cdots & x_{n,2} \\ \vdots & \vdots & \ddots & \vdots \\ x_{1,n} & x_{2,n} & \cdots & x_{n,n} \end{bmatrix} \begin{bmatrix} t_1 \\ t_2 \\ \vdots \\ t_n \end{bmatrix}$$
$$\text{s.t.} \quad \mathbf{b}^T \mathbf{t} = t_1 + t_2 + \cdots + t_n = 1,$$
$$\mathbf{t} \equiv [t_1 \ t_2 \ \cdots \ t_n]^T,$$
$$\mathbf{b} \equiv [1 \ 1 \ \cdots \ 1]^T.$$

From the minimum-distance-path perspective, the distance between $\mathbf{P}$ and the plane spanned by $\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_n$ is

$$d = \min \|\mathbf{P}(\mathbf{t}) - \mathbf{P}\|_2 = \min \|\mathbf{C} \mathbf{t} - \mathbf{P}\|_2 = \min \sqrt{(\mathbf{C} \mathbf{t} - \mathbf{P})^T (\mathbf{C} \mathbf{t} - \mathbf{P})}$$
$$\text{s.t.} \quad \mathbf{b}^T \mathbf{t} = 1.$$

$$\mathbf{t} = \mathbf{C}^{-1} [\mathbf{P} + \frac{(1 - \bar{\mathbf{b}}^T \mathbf{P})}{\bar{\mathbf{b}}^T \bar{\mathbf{b}}} \bar{\mathbf{b}}]$$

$$\bar{\mathbf{b}} \equiv \mathbf{C}^{-T} \mathbf{b}$$

Distance

- Point-To-Plane Distance

Any point on the plane spanned by $\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_n$ can be parametrized as

$$\mathbf{P}(\mathbf{t}) = \mathbf{C} \mathbf{t} \equiv [\mathbf{P}_1 \ \mathbf{P}_2 \ \cdots \ \mathbf{P}_n] \mathbf{t} \equiv \begin{bmatrix} x_{1,1} & x_{2,1} & \cdots & x_{n,1} \\ x_{1,2} & x_{2,2} & \cdots & x_{n,2} \\ \vdots & \vdots & \ddots & \vdots \\ x_{1,n} & x_{2,n} & \cdots & x_{n,n} \end{bmatrix} \begin{bmatrix} t_1 \\ t_2 \\ \vdots \\ t_n \end{bmatrix}$$

s.t. $\mathbf{b}^T \mathbf{t} = t_1 + t_2 + \cdots + t_n = 1,$

$$\mathbf{t} \equiv [t_1 \ t_2 \ \cdots \ t_n]^T,$$
$$\mathbf{b} \equiv [1 \ 1 \ \cdots \ 1]^T.$$

$$\mathbf{t} = \mathbf{C}^{-1} [\mathbf{P} + \frac{(1 - \bar{\mathbf{b}}^T \mathbf{P})}{\bar{\mathbf{b}}^T \bar{\mathbf{b}}} \bar{\mathbf{b}}] \quad \bar{\mathbf{b}} \equiv \mathbf{C}^{-T} \mathbf{b}$$

The projection of $\mathbf{P}$ on the plane spanned by $\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_n$ is

$$\mathbf{P}_{proj} = \mathbf{C} \mathbf{C}^{-1} [\mathbf{P} + \frac{(1 - \bar{\mathbf{b}}^T \mathbf{P})}{\bar{\mathbf{b}}^T \bar{\mathbf{b}}} \bar{\mathbf{b}}] = \mathbf{P} + \frac{(1 - \bar{\mathbf{b}}^T \mathbf{P})}{\bar{\mathbf{b}}^T \bar{\mathbf{b}}} \bar{\mathbf{b}}.$$

The distance between $\mathbf{P}$ and the plane spanned by $\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_n$ is

$$d = \sqrt{(\mathbf{P}_{proj} - \mathbf{P})^T (\mathbf{P}_{proj} - \mathbf{P})}.$$

Topological Relationship

- Point-Plane Topological Relationship

To judge whether a given point $\mathbf{P}$ is located on the plane spanned by n given points

$$\begin{aligned}\mathbf{P}_{proj} = \mathbf{P} &\iff \mathbf{P} + \frac{(1 - \bar{\mathbf{b}}^T \mathbf{P})}{\bar{\mathbf{b}}^T \bar{\mathbf{b}}} \bar{\mathbf{b}} = \mathbf{P} \iff \frac{(1 - \bar{\mathbf{b}}^T \mathbf{P})}{\bar{\mathbf{b}}^T \bar{\mathbf{b}}} \bar{\mathbf{b}} = \mathbf{0} \\ &\iff 1 - \bar{\mathbf{b}}^T \mathbf{P} = 0 \iff \mathbf{b}^T \mathbf{C}^{-1} \mathbf{P} = 1.\end{aligned}$$

In other words, compute the coefficients of $\mathbf{P}$ expressed linearly in terms of $\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_n$

$$\mathbf{t}_P = \mathbf{C}^{-1} \mathbf{P}$$

and then check whether the summation of the elements of $\mathbf{t}_P$ is equal to 1.

Degenerate case ($n=3$ as example)

$$\mathbf{t}_P \equiv \begin{bmatrix} t_1 \\ t_2 \\ t_3 \end{bmatrix} = \begin{bmatrix} \mathbf{P}_1 & \mathbf{P}_2 & \mathbf{P}_3 \\ 1 & 1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{P} \\ 1 \end{bmatrix}$$

- Linear Equation Solving
- $\mathbf{p}\mathbf{v} = \mathbf{A} \setminus \mathbf{B}$

```
% p + 30 v = 30.3
% p + 35 v = 34.9
% p + 40 v = 40.4
% p + 45 v = 45.1
% p + 50 v = 49.5
% p + 55 v = 55.9
% p + 60 v = 59.8
```


Topological Relationship

- Planar Point-Triangle Topological Relationship

The method can also be adapted to judge on a two-dimensional plane whether a given point is inside a given triangle. Given four points $\mathbf{P}_1$, $\mathbf{P}_2$, $\mathbf{P}_3$, and $\mathbf{P}$ on the two-dimensional plane

$$\mathbf{P}_1 = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}, \quad \mathbf{P}_2 = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}, \quad \mathbf{P}_3 = \begin{bmatrix} x_3 \\ y_3 \end{bmatrix}, \quad \mathbf{P} = \begin{bmatrix} x \\ y \end{bmatrix}.$$

and suppose $\mathbf{P}_1$, $\mathbf{P}_2$, $\mathbf{P}_3$ are not colinear.

$$\mathbf{P}_1^A = \begin{bmatrix} x_1 \\ y_1 \\ 1 \end{bmatrix}, \quad \mathbf{P}_2^A = \begin{bmatrix} x_2 \\ y_2 \\ 1 \end{bmatrix}, \quad \mathbf{P}_3^A = \begin{bmatrix} x_3 \\ y_3 \\ 1 \end{bmatrix}, \quad \mathbf{P}^A = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}.$$

Compute the coefficients t_1 , t_2 , t_3 of $\mathbf{P}^A$ expressed linearly in terms of $\mathbf{P}_1^A$, $\mathbf{P}_2^A$, $\mathbf{P}_3^A$

$$\begin{aligned} [\mathbf{P}_1^A \quad \mathbf{P}_2^A \quad \mathbf{P}_3^A] \begin{bmatrix} t_1 \\ t_2 \\ t_3 \end{bmatrix} &= \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} t_1 \\ t_2 \\ t_3 \end{bmatrix} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} t_1 \\ t_2 \\ t_3 \end{bmatrix} &= [\mathbf{P}_1^A \quad \mathbf{P}_2^A \quad \mathbf{P}_3^A]^{-1} \mathbf{P}^A = \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ 1 & 1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}. \end{aligned}$$

Check

$$t_1 \geq 0, \quad t_2 \geq 0, \quad t_3 \geq 0$$

Topological Relationship

- Planar Point-Triangle Topological Relationship

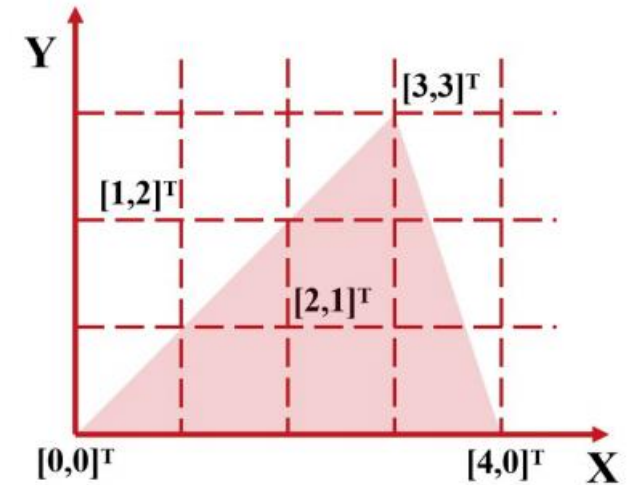
For example, consider three points $\mathbf{P}_1, \mathbf{P}_2, \mathbf{P}_3$

$$\mathbf{P}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{P}_2 = \begin{bmatrix} 4 \\ 0 \end{bmatrix}, \quad \mathbf{P}_3 = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

which span a triangle, as illustrated in Figure 3.2. Judge whether two extra points $\mathbf{P}_4$ and $\mathbf{P}_5$

$$\mathbf{P}_4 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{P}_5 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

are inside the triangle spanned by $\mathbf{P}_1, \mathbf{P}_2, \mathbf{P}_3$ or not.



Topological Relationship

- Planar Point-Triangle Topological Relationship

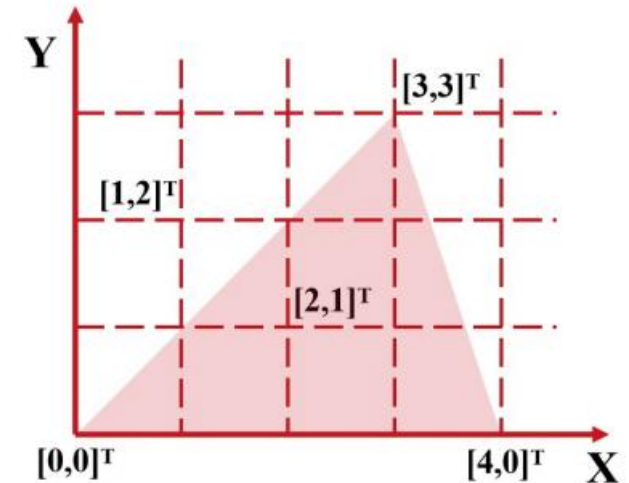
First, augment these points to

$$\mathbf{P}_1^A = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{P}_2^A = \begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{P}_3^A = \begin{bmatrix} 3 \\ 3 \\ 1 \end{bmatrix}, \quad \mathbf{P}_4^A = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{P}_5^A = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}.$$

Compute

$$\mathbf{t}_{\mathbf{P}_4} = [\mathbf{P}_1^A \quad \mathbf{P}_2^A \quad \mathbf{P}_3^A]^{-1} \mathbf{P}_4^A = \begin{bmatrix} 0 & 4 & 3 \\ 0 & 0 & 3 \\ 1 & 1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 7/12 \\ -1/4 \\ 2/3 \end{bmatrix},$$

$$\mathbf{t}_{\mathbf{P}_5} = [\mathbf{P}_1^A \quad \mathbf{P}_2^A \quad \mathbf{P}_3^A]^{-1} \mathbf{P}_5^A = \begin{bmatrix} 0 & 4 & 3 \\ 0 & 0 & 3 \\ 1 & 1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5/12 \\ 1/4 \\ 1/3 \end{bmatrix}.$$



Generalized Distance

- Mahalanobis Distance

$$d_{\mathbf{M}}(\mathbf{P}_1, \mathbf{P}_2) \equiv \|\mathbf{P}_1 - \mathbf{P}_2\|_{\mathbf{M}} \equiv \sqrt{(\mathbf{P}_1 - \mathbf{P}_2)^T \mathbf{M}^{-1} (\mathbf{P}_1 - \mathbf{P}_2)}$$

$$M = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix} \quad P_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad P_2 = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

$$d_M(P_1, P_2) = \sqrt{\left(\begin{bmatrix} -1 \\ 1 \end{bmatrix} - \begin{bmatrix} 3 \\ 5 \end{bmatrix} \right)^T \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}^{-1} \left(\begin{bmatrix} -1 \\ 1 \end{bmatrix} - \begin{bmatrix} 3 \\ 5 \end{bmatrix} \right)} = \sqrt{\frac{(-1 - 3)^2}{1} + \frac{(1 - 5)^2}{4}} = \sqrt{20}$$

Generalized Distance

- City-Block Distance

The city-block distance between two two-dimensional points

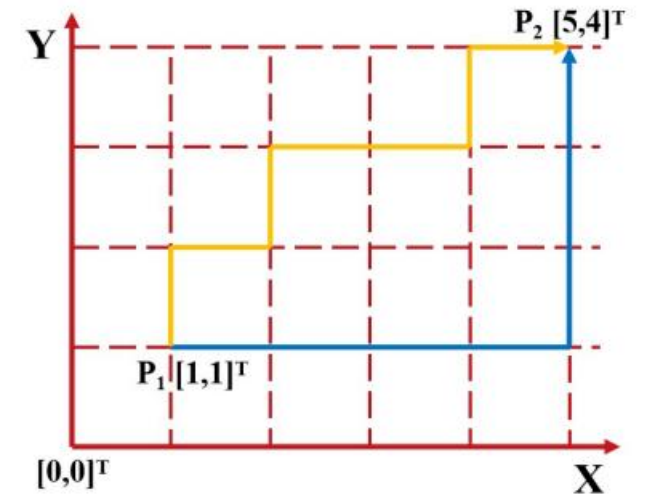
$$\mathbf{P}_1 = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}, \quad \mathbf{P}_2 = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix},$$

is right their L_1 -norm distance on the two-dimensional plane

$$\|\mathbf{P}_1 - \mathbf{P}_2\|_1 = |x_1 - x_2| + |y_1 - y_2|.$$

$$P_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad P_2 = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$$

$$\|\mathbf{P}_2 - \mathbf{P}_1\|_1 = |5 - 1| + |4 - 1| = 4 + 3 = 7$$



Generalized Distance

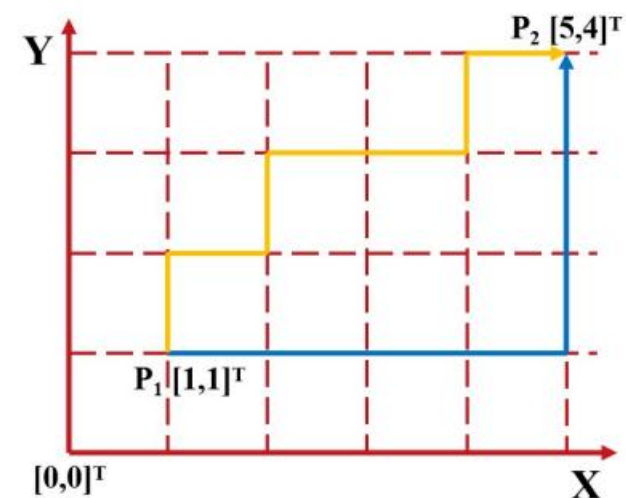
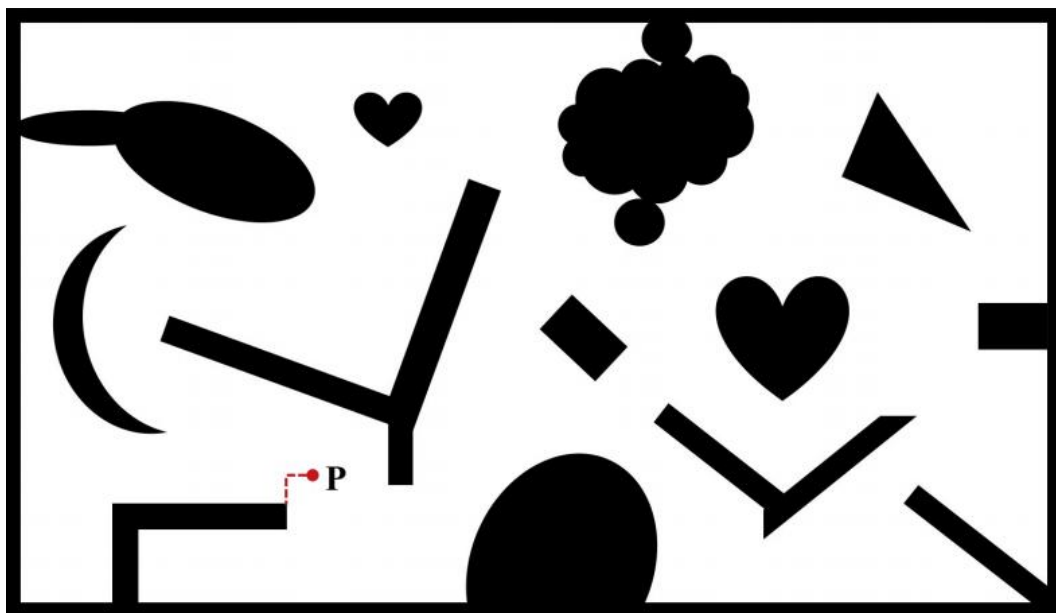
- City-Block Distance

The city-block distance between two two-dimensional points

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is right their L_1 -norm distance on the two-dimensional plane

$$\|\mathbf{P}_1 - \mathbf{P}_2\|_1 = |x_1 - x_2| + |y_1 - y_2|.$$



Generalized Distance

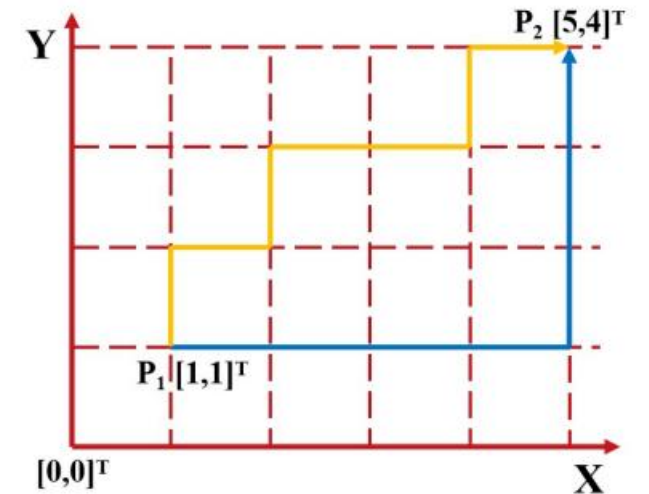
- Obstacle Distance Map

2	1	1	2	3	2	3	2	1
1	0	0	1	2	1	2	1	0
2	1	1	2	1	0	1	2	1
3	2	2	2	1	0	1	2	2
4	3	3	3	2	1	2	3	3

$$I(u, v) := \min\{1 + I(u, v - 1), 1 + I(u - 1, v), I(u, v)\}$$

$$I(u, v) := \min\{1 + I(u, v + 1), 1 + I(u + 1, v), I(u, v)\}$$

$$\|\mathbf{P}_1 - \mathbf{P}_2\|_1 = |x_1 - x_2| + |y_1 - y_2|.$$

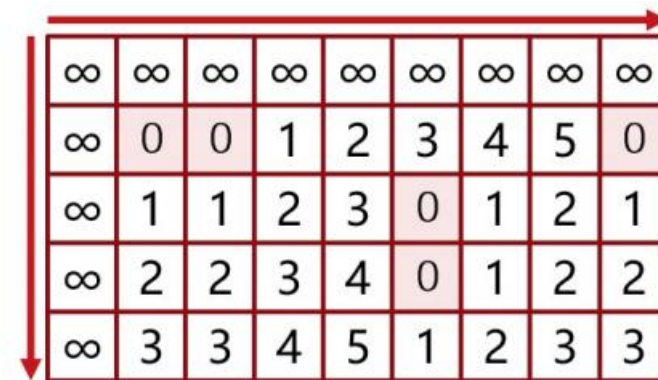
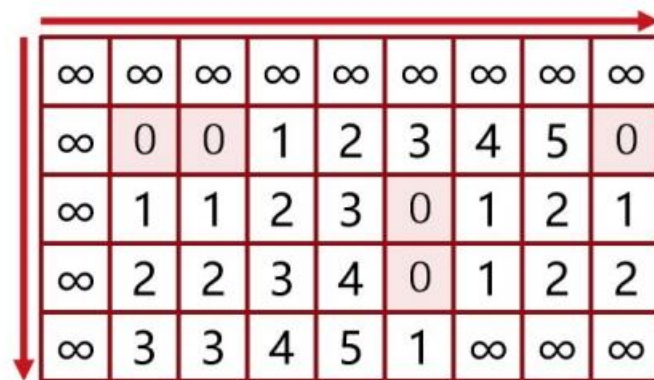
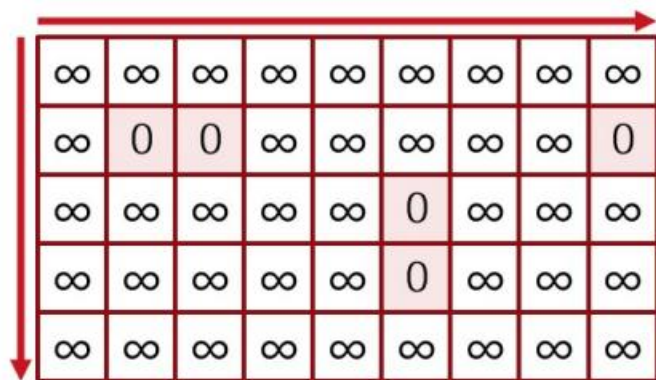
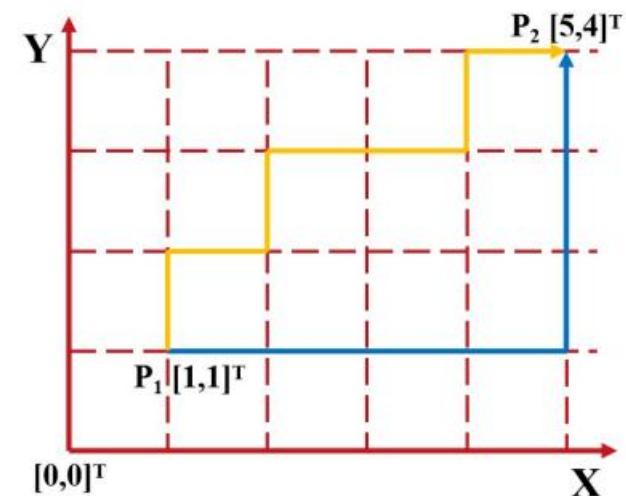


Generalized Distance

- Obstacle Distance Map

$$I(u, v) := \min\{1 + I(u, v - 1), 1 + I(u - 1, v), I(u, v)\}$$

$$\|P_1 - P_2\|_1 = |x_1 - x_2| + |y_1 - y_2|.$$

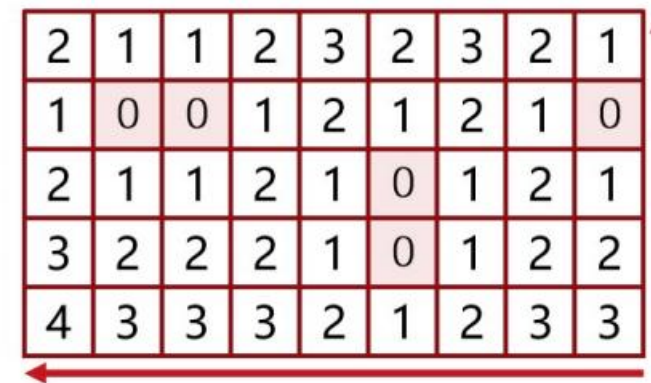
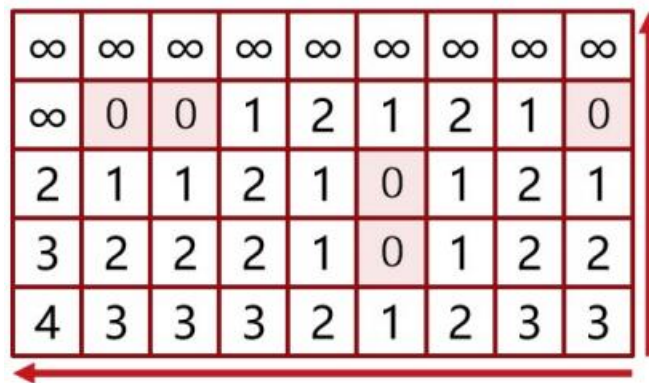
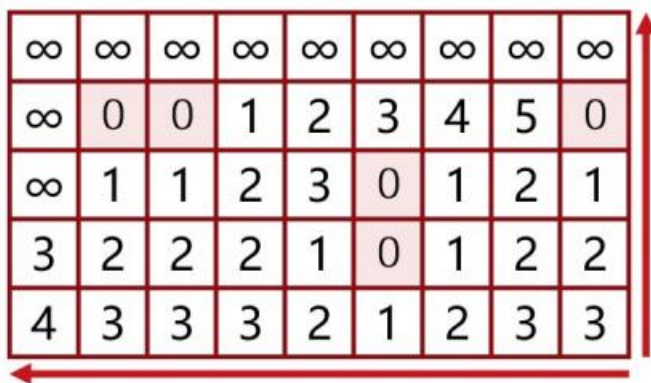
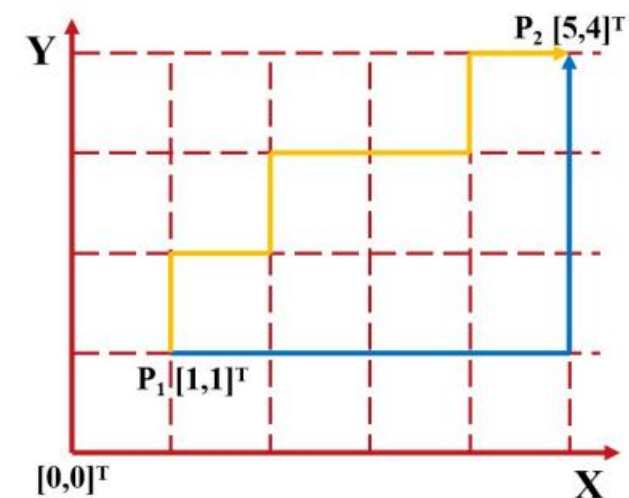


Generalized Distance

- Obstacle Distance Map

$$I(u, v) := \min\{1 + I(u, v + 1), 1 + I(u + 1, v), I(u, v)\}$$

$$\|P_1 - P_2\|_1 = |x_1 - x_2| + |y_1 - y_2|.$$

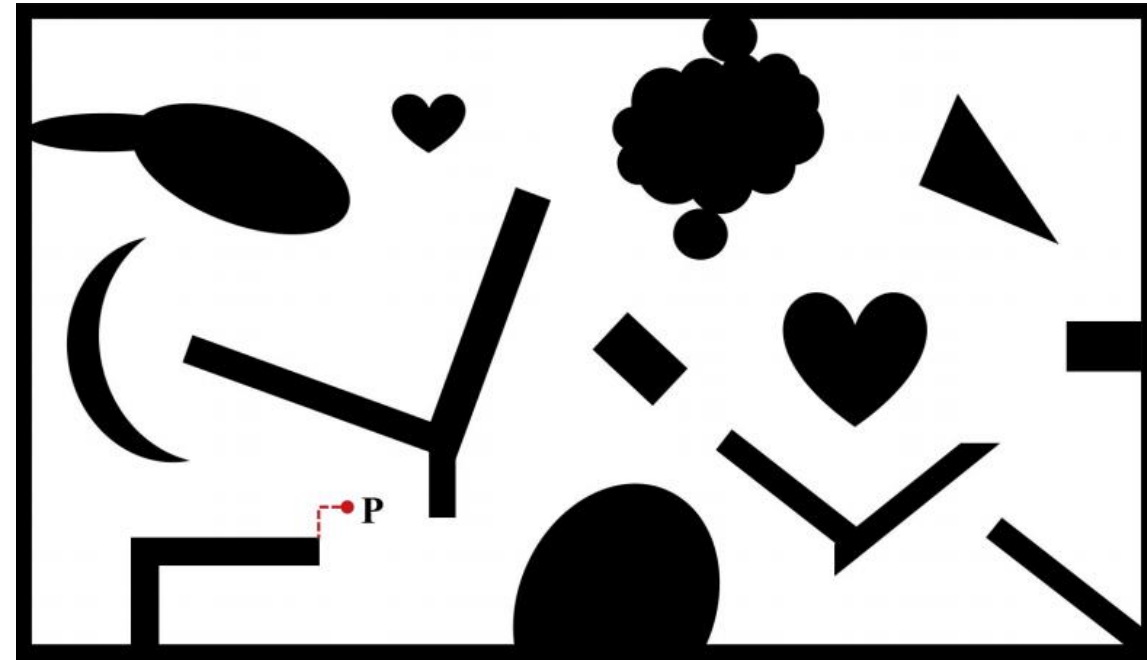


Generalized Distance

- Obstacle Distance Map

$$I(u, v) := \min\{1 + I(u, v - 1), 1 + I(u - 1, v), I(u, v)\}$$

$$I(u, v) := \min\{1 + I(u, v + 1), 1 + I(u + 1, v), I(u, v)\}$$



DISCUSSION

Q & A



Graph

- History

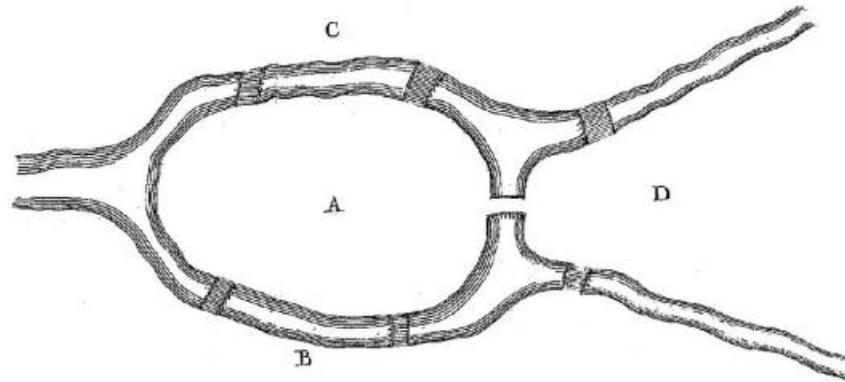
- *Leonhard **EULER***

- “Solutio problematis ad geometriam situs pertinentis”, 1736

- means “solution of the problem (with regard) to the geometry of a connected region”

- Debut of graph theory

- *Seven bridges of Konigsberg* (Kaliningrad of Russia, hometown of *Emmanuel Kant*)



Graph

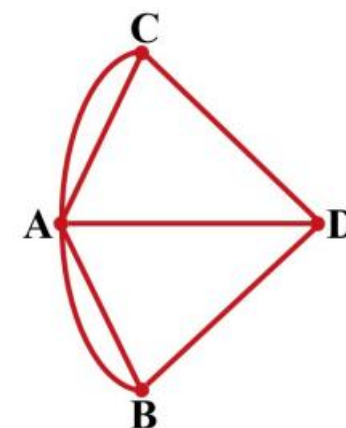
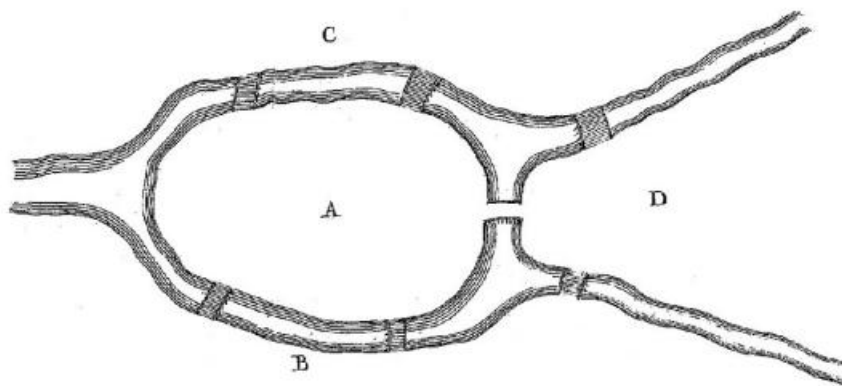
- Topological Abstraction

- Vertices

$$V = \{V_1, V_2, \dots, V_{|V|}\}$$

- Edges

$$E = \{E_1, E_2, \dots, E_{|E|}\}$$



Graph

- Topological Abstraction

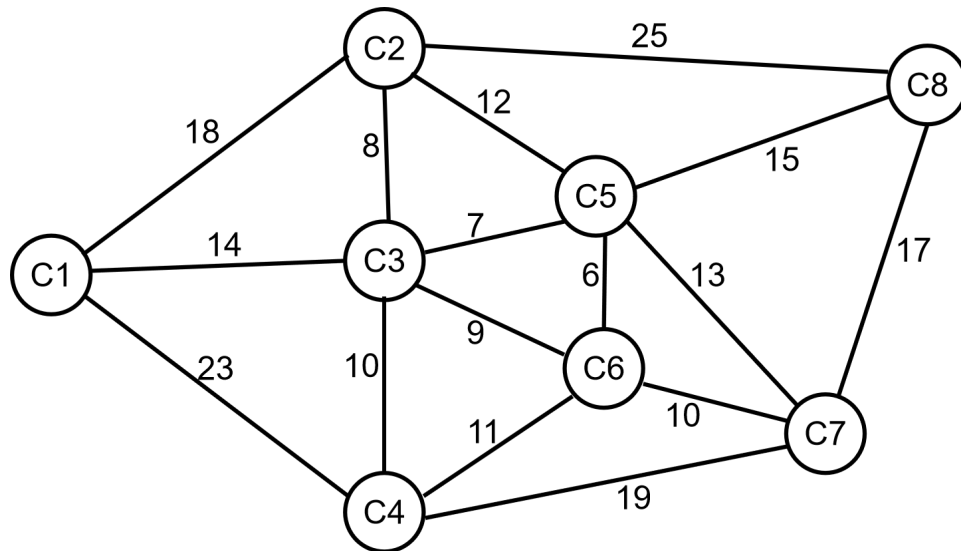
- Vertices

$$\mathbf{V} = \{V_1, V_2, \dots, V_{|\mathbf{V}|}\}$$

- Edges

$$\mathbf{E} = \{E_1, E_2, \dots, E_{|\mathbf{E}|}\}$$

- Adjacency matrix



	C1	C2	C3	C4	C5	C6	C7	C8
C1	0	18	14	23	0	0	0	0
C2	18	0	8	0	12	0	0	25
C3	14	8	0	10	7	9	0	0
C4	23	0	10	0	0	11	19	0
C5	0	12	7	0	0	6	13	15
C6	0	0	9	11	6	0	10	0
C7	0	0	0	19	13	10	0	17
C8	0	25	0	0	15	0	17	0

Graph

- Topological Abstraction

- Vertices

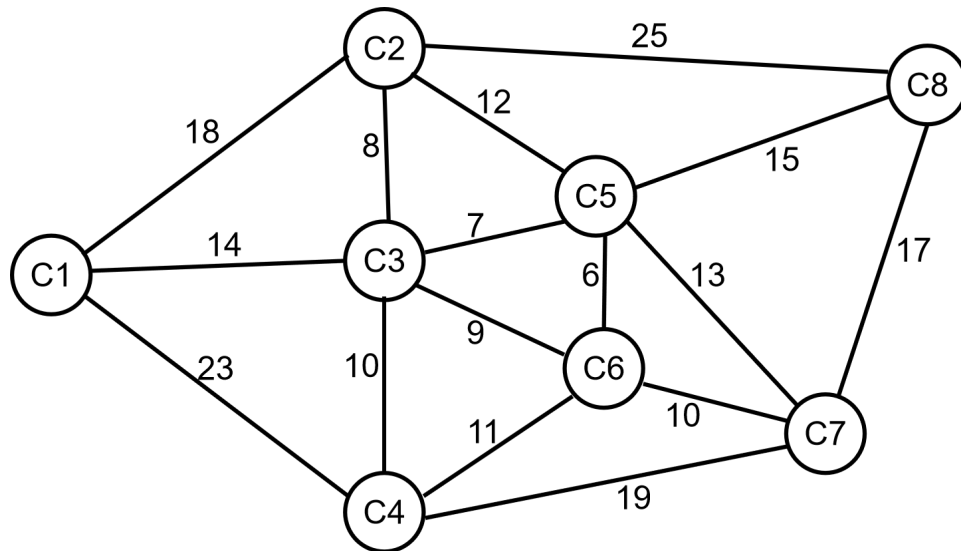
$$\mathbf{V} = \{V_1, V_2, \dots, V_{|\mathbf{V}|}\}$$

- Edges

$$\mathbf{E} = \{E_1, E_2, \dots, E_{|\mathbf{E}|}\}$$

- Adjacency matrix

- Adjacency list



C1	C2	18	C3	14	C4	23				
C2	C1	18	C3	8	C5	12	C8	25		
C3	C1	14	C2	8	C4	10	C5	7	C6	9
C4	C1	23	C3	10	C6	11	C7	19		
C5	C2	12	C3	7	C6	6	C7	13	C8	15
C6	C3	9	C4	11	C5	6	C7	10		
C7	C4	19	C5	13	C6	10	C8	17		
C8	C2	25	C5	15	C7	17				



Convex Geometry

- Convex Set and Convex Hull

Convex sets are important geometric entities that play an indispensable role in computational geometry. A convex set $\mathbf{C}$ is a set such that for any pair of points $\mathbf{P}_1, \mathbf{P}_2 \in \mathbf{C}$ and any t satisfying $0 \leq t \leq 1$, one has

$$t\mathbf{P}_1 + (1 - t)\mathbf{P}_2 \in \mathbf{C},$$

or equivalently speaking, is a set such that for any group of points $\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_k \in \mathbf{C}$ and any coefficients $t_1, t_2, \dots, t_k$ satisfying

$$t_1 \geq 0, \quad t_2 \geq 0, \quad \dots, \quad t_k \geq 0, \quad t_1 + t_2 + \dots + t_k = 1,$$

one has

$$t_1\mathbf{P}_1 + t_2\mathbf{P}_2 + \dots + t_k\mathbf{P}_k \in \mathbf{C}.$$

The linear combination of points $\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_k$ is called their **convex combination**.

Given an arbitrary set $\mathbf{S}$ (not necessarily convex), the set of all convex combinations of points in $\mathbf{S}$ namely

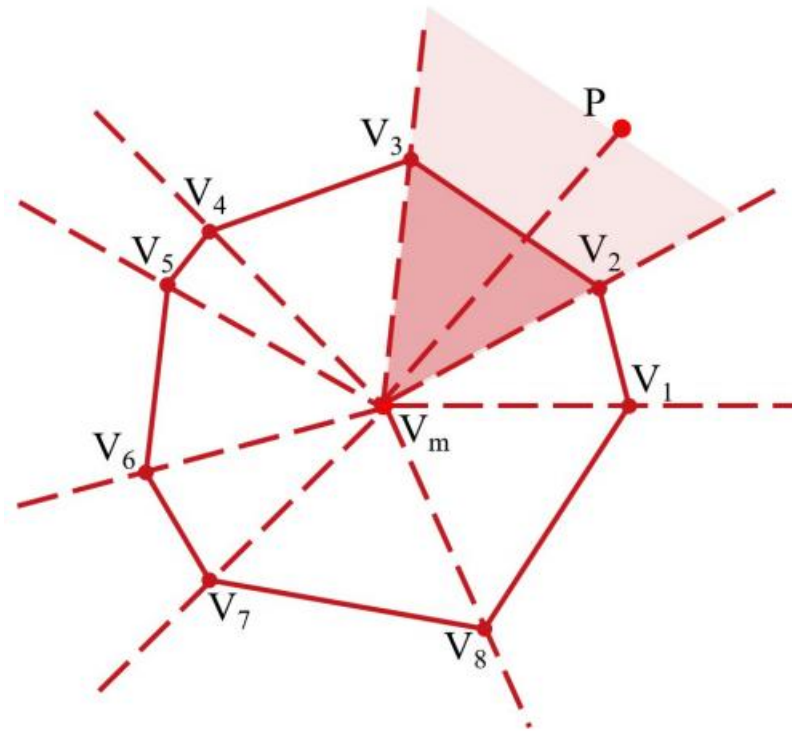
$$\mathbf{conv} \mathbf{S} \equiv \{t_1\mathbf{P}_1 + \dots + t_k\mathbf{P}_k \mid \forall k, \mathbf{P}_i \in \mathbf{S}, t_i \geq 0, i = 1, \dots, k; \sum_{i=1}^k t_i = 1\}$$

is called the **convex hull** of $\mathbf{S}$.

The convex hull $\mathbf{conv} \mathbf{S}$ is the minimum convex set that contains $\mathbf{S}$.

Convex Geometry

- Convex Polygon



DISCUSSION

Q & A

